Synthesis of High-Fidelity Fluorescent Cell Imagery via a Self-Developed Diffusion Model

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Abstract

This paper presents a method for synthesising fluorescent cell images using a Denoising Diffusion Probabilistic Model (DDPM) aimed at advancing the field of biomedical imaging and neuroscience. Building on the DDPM framework, I convert greyscale, channel-separated fluorescent microscopy data into composite colour images and apply random cropping to enhance variability. The model begins by employing a forward diffusion process to add Gaussian noise to an image at a rate specified by the noise schedule. The model then learns to reverse the Gaussian noise process to recover key cellular features, and while the outputs still show artefacts, preliminary results indicate that essential morphological patterns are emerging. Additionally, I discuss hardware and ethical considerations and highlight the need for more accessible educational resources on diffusion models. Overall, this work lays a foundation for augmenting limited biomedical datasets and advancing diagnostic imaging.

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2. Introduction

The rapid evolution of generative models has led to significant interest in the synthesis of high-fidelity images across various domains, including biomedical imaging. In recent years, diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPMs), have emerged as a promising approach for image synthesis. Originally proposed by Sohl-Dickstein et al. [1] and further refined by Ho et al. [2] as well as Nichol and Dhariwal [3], these models utilise a process inspired by nonequilibrium thermodynamics, progressively corrupting an image with Gaussian noise and then learning to reverse the process through a neural network. This iterative denoising mechanism not only captures the complex data distribution but also offers enhanced training stability and mode coverage when compared to other generative frameworks such as Generative Adversarial Networks (GANs) [4].

In this paper, I explore the application of DDPMs to the generation of synthetic fluorescent cell images, a task of particular relevance to biomedical imaging and neuroscience. Fluorescent microscopy is a crucial technique for visualising cellular structures and monitoring dynamic biological processes. However, experimental limitations such as photobleaching, phototoxicity, and the constrained availability of high-quality and large datasets often impede progress in this area [5]. By employing DDPMs, my first objective is to develop and train a model which can synthesise images that preserve the structural nuances and biological fidelity of real fluorescent cell images, thereby augmenting limited datasets and providing a foundation for subsequent analytical applications. Secondly, I aim to develop the model in a manner that is approachable for beginners, thereby improving people's understanding and curbing misinformation surrounding this field.

In my work I undertook a rigorous data preparation process to ensure that the input images sourced from the BioImage Archive were accurately pre-processed to meet the requirements of the project. Additionally, ethical considerations, including responsible use and data integrity, have been paramount throughout the project. Although the images generated show the model is not yet ready for deployment in medical research due to the presence of artefacts and limited validation metrics, initial results indicate that it is beginning to learn fundamental cellular patterns.

3. Background

The recent surge in popularity and development of generative modelling techniques has positioned diffusion models to be at the forefront of a class of algorithms promising high-fidelity synthetically generated images. Diffusion models work by gradually adding Gaussian noise to an image, corrupting it and training the model to learn the amount of noise added, thereby allowing it to reverse this corruption and generate new images. This technique has given rise to groundbreaking results, offering unparalleled performance and accuracy in domains such as natural images, complex artwork, medical scans [6], and, as researched in this paper, fluorescent cell imagery.

3.1 Outline of Diffusion Models

Denoising Diffusion Probabilistic Models, or DDPMs, are a class of diffusion models that were originally proposed by Ho et al. [2]. Their framework is based upon the work by Sohl-Dickstein et al., where a Markov Chain is used to add Gaussian noise to an image over a certain number of timesteps T. In the paper, they define this formula as:

$$q(x_{1:T} \mid x_0) \coloneqq \prod_{t=1}^{T} q(x_t \mid x_{t-1}), \qquad q(x_t \mid x_{t-1}) \coloneqq \mathcal{N}\left(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\right)$$

where q is the forward diffusion function, x_0 is the original image represented as a tensor, t is the current timestep, x_t is the final noisy image, β_t is the amount of noise added at a timestep t, I is an identity matrix and $\mathcal{N}$ is the Gaussian distribution being modelled.

The reverse diffusion also works on a Markov Chain, reversing the learned Gaussian distribution to generate new images using the formula:

$$p_\theta(x_{0:T}) \coloneqq p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), p_\theta(x_{t-1} \mid x_t) \coloneqq \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

where p_θ is the learned probability distribution of the amount of noise added to an image, μ_θ is the mean of the conditional Gaussian distribution, and Σ_θ is the covariance of the same conditional Gaussian distribution.

In the reverse diffusion phase, the model begins with a sample of pure noise and iteratively denoises the image over multiple timesteps until a near-original version is recovered. At each iteration, the neural network predicts the residual noise present in the current image, which is then subtracted to progressively remove the noise layer by layer. Crucially, this process relies on a noise schedule that dictates how much noise is added or removed at each timestep. By ensuring that early steps remove coarse noise while later steps focus on fine details, the model preserves essential structural and contextual information. Techniques such as adaptive scaling during the denoising steps help to maintain a balance between removing noise and retaining critical features inherent in the original data.

Researchers have shown that, with appropriate noise scheduling and network architectures, these models can produce images that rival or even surpass the quality of those generated by GANs. Unlike GANs, however, diffusion models exhibit superior mode coverage and training stability, making them particularly attractive for applications in sensitive fields like biomedical imaging [4].

3.2 Criticism of Current Learning Resources for Diffusion Models

Despite the remarkable progress in diffusion models over the past few years, I find the learning and teaching material available in this field to be inaccessible to beginners. Many of the seminal papers, such as those by Ho et al. and Nichol and Dhariwal present these models through highly mathematical formulations and advanced stochastic concepts. While their works are invaluable in

advancing the field of generative modelling, they are written in a manner that is challenging for newcomers and even intermediate learners to parse.

Most existing tutorials and online resources tend to focus on experimental results and intricate theoretical derivations without providing sufficient context or step-by-step explanations. Essential aspects, including noise scheduling, reverse diffusion processes, and the UNet architecture, are frequently presented with insufficient detail. This fragmented approach leaves learners to piece together the entire picture from multiple sources, hindering clear understanding and practical implementation. Simple fixes could involve explaining the "why?" behind concepts and presenting the reader with small-scale worked out examples of the formula, showing what the model does behind the scenes.

One of my key motivations for this project is to bridge this educational gap by designing a diffusion model that is much simpler than existing works, based closely on the core theory of DDPMs. My goal is to develop a model that is easier to read, parse, and understand, without greatly sacrificing performance.

Simplifying the model and providing clearer, more accessible explanations can significantly lower the barrier to entry, allowing a broader range of researchers and students to engage with and build upon diffusion-based methodologies. This approach not only encourages wider adoption but also facilitates more rapid innovation, as clearer instructional material fosters better comprehension and more effective troubleshooting.

3.3 Current Literature Review

The explosive growth in popularity of diffusion models has seen it change drastically from its inception 10 years ago, by Sohl-DickStein et al. [1]. Their proposal employed techniques from the diffusion field of non-equilibrium thermodynamics, where one form of distribution is converted into another [7]. In their paper, a data sample is gradually corrupted by adding small amounts of Gaussian noise in a Markov chain until it becomes nearly pure noise. The reverse process then learns to generate new samples by iteratively removing this noise. While groundbreaking at the time, their approach involved a complex training objective that proved challenging for optimisation and scalability, which limited its practical applicability.

5 years later, Ho et al. significantly refined this framework by simplifying the reverse diffusion process such that the network was tasked with directly predicting the amount of noise added at each timestep. This reframing allowed the adoption of a straightforward Mean Squared Error (MSE) loss function between the true noise added and the network's prediction. By focusing on noise prediction, the model could leverage well-established regression techniques, thereby stabilizing training and improving convergence.

Moreover, they introduced practical adjustments to the noise schedule, further enhancing the quality and diversity of the generated images. Their modifications led to substantial improvements in sample fidelity and diversity, enabling diffusion models to outperform other generative frameworks such as GANs in several image generation tasks. Overall, their contributions

transformed diffusion models from a theoretical, seemingly utopian idea first proposed by Sohl-Dickstein et al., into a robust and scalable approach for high-quality image generation in various domains.

Nichol and Dhariwal built upon the work established by Ho et al. by introducing several practical enhancements that significantly improved the quality, efficiency, and scalability of DDPMs. They proposed a new noise schedule based on a cosine function, which provided a smoother decay of signal-to-noise ratio over timesteps, thereby addressing instabilities in the gradient dynamics observed with linear schedules that were used by Ho et al. This adjustment not only simplified the training process but also led to a more balanced learning across all timesteps. For these reasons, this is the noise schedule I employed in the development of my model.

In addition, Nichol and Dhariwal introduced modifications to the network architecture and the reverse process. They reparametrized the model's denoising objective, demonstrating that predicting the noise residual directly, when coupled with a reweighted mean-squared error loss, yielded superior sample quality, as evidenced by improved Fréchet Inception Distance (FID) scores. Their extensive studies revealed that these architectural and objective function modifications reduced the need for excessively long diffusion chains, thus enabling high-fidelity image synthesis with fewer reverse steps and, consequently, more efficient inference.

Furthermore, their work also explored techniques for conditional generation, such as classifier-free guidance, which allowed for improved control over the synthesis process without sacrificing the overall stability of training. Collectively, the contributions of Nichol and Dhariwal represent a notable advancement in the practical application of diffusion models, establishing them as a competitive alternative to other generative models, such as GANs, for high-resolution image synthesis in diverse domains.

Consequently, the application of these models in medical fields such as synthetic medical image generation, noise removal and inpainting amongst others has shown promising results in realism and preservation of fine anatomical details.

Kazerouni et al. provided a comprehensive survey of the existing literature and works in the application of diffusion models in the field of medical imaging. Unlike GANs and Variational Autoencoders (VAEs), they show that diffusion models provide a greater sampling quality, better mode coverage and resistance to "mode collapse, vanishing gradients and convergence." [8].

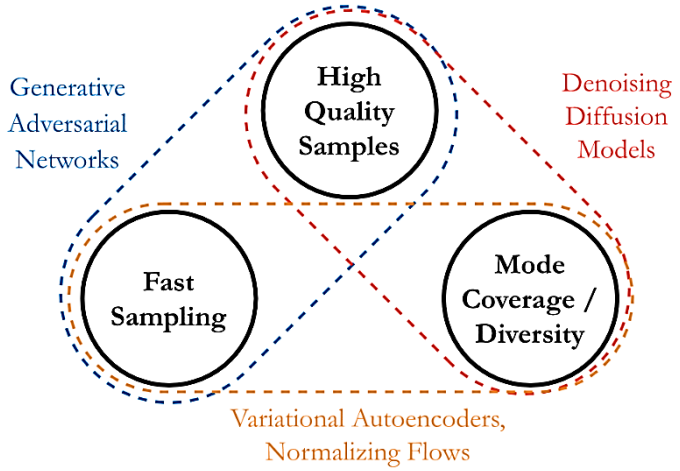

Fig. 1. Generative learning trilemma [8]

Khader et al. demonstrated the application of diffusion models in synthesizing high quality images of Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) scans. Their research showed that diffusion models achieve a higher Multi-Scale Structural Similarity Index Measure (MS-SSIM) score when compared to GANs, showing that "diffusion models can generate more diverse samples representative of the original data distribution." They also showed that synthetically generated data can be used to train neural networks, demonstrating the potential of eliminating the need to obtain real images to train diffusion models. However, this study was done using a relatively small dataset, with the generated 3D images not having the standard diagnostic resolution. This is proposed as a key area of focus for future work, as they found a direct correlation between compression in the latent space and accuracy of generated images [4].

Dorjsembe et al. proposed a complementary approach using conditional diffusion models for medical image synthesis. Their framework extends the DDPM architecture with semantic conditioning, allowing the model to generate images aligned with specific clinical labels, such as presence of a tumour, organ type, or pathology class. Applied to MRI and CT data, their method showed strong results in generating class-consistent and anatomically realistic images. Crucially, they demonstrated that models trained on these synthetic images improved performance on downstream segmentation tasks, particularly in cases with limited annotated real data. The authors also discussed the privacy preserving benefits of synthetic data generation, arguing that conditional diffusion could provide a viable pathway for sharing medical data in research while maintaining patient anonymity [9].

Friedrich et al. introduced Wavelet Diffusion Models (WDMs), a new approach that combines the strengths of diffusion models with the multi-scale representation of wavelet transforms. In this architecture, the diffusion process is applied not in the pixel space, but in the wavelet domain, where images are decomposed into spatial-frequency components. This enables the model to generate images by progressively denoising features at different spatial scales, from coarse global

structures down to fine local details. In the context of medical imaging, this is particularly valuable: anatomical structures like organs, vessels, and lesions exist across different frequency bands, and a multi-scale generative approach ensures sharper boundaries and more coherent textures. WDM significantly reduces the computational load of high-resolution image synthesis, as most of the generation occurs in a lower-dimensional representation before wavelet-based reconstruction. When tested on various medical datasets, including retinal fundus images and chest X-rays, WDM outperformed baseline DDPMs in image quality metrics (e.g. FID, SSIM) and was found to produce more anatomically plausible samples [10].

Together, these works highlight the growing sophistication of diffusion-based approaches in medical image synthesis. By leveraging frequency-domain modelling of WDM, conditional sampling strategies, and multi-dimensional representations, recent research is pushing diffusion models beyond general-purpose generative tools into tailored frameworks that can address the specific structural, computational, and ethical challenges of modern medical imaging.

3.4 Rationale for Choosing Fluorescent Cell Imagery

Fluorescent cell imaging is a foundational practice in neuroscience and biomedical research, allowing us to reveal intricate cellular and subcellular structures [11]. Here, fluorescent probes are either chemically synthesized or genetically encoded, and bind selectively to target biomolecules, enabling researchers to visualise specific proteins, organelles, or nucleic acids with high contrast and specificity [12]. Techniques such as confocal and two-photon microscopy have been widely used to visualise the spatial organisation and dynamic behaviour of neural circuits [13]. Notably, Yuste and Bonhoeffer demonstrated that morphological changes in dendritic spines, observable via fluorescent labelling, are closely associated with synaptic plasticity and learning processes [14]. This fundamental insight underscores the value of fluorescent imaging for understanding the structural correlates of neural function.

A major motivation for selecting fluorescent cell images in this study is their capacity to capture clinically and experimentally relevant details. Fluorescent imaging enables the observation of subtle variations in protein expression, ion channel distribution, and signalling pathways that are critical in both healthy neural processes and pathological conditions such as neurodegenerative diseases [15]. Moreover, time-lapse fluorescence microscopy provides a dynamic view of cellular events, making it possible to study processes like synaptic formation, migration of neural stem cells, and intracellular trafficking in real time; although this suffers from challenges similar to those discussed below [16]. These capabilities are essential not only for neuroscience research but also for advancing diagnostic methods and therapeutic interventions.

However, fluorescent microscopy is not without its challenges. Photobleaching, which is the irreversible loss of fluorescence due to prolonged light exposure is a persistent issue that can compromise image quality [17]. Phototoxicity may alter cell physiology or viability during live-cell imaging [18]. In addition, autofluorescence from endogenous biomolecules frequently introduces background noise, reducing the signal-to-noise ratio and complicating data interpretation [19]. These limitations contribute to the often limited size and substandard quality

of fluorescent cell imaging datasets, posing significant obstacles for research, data-driven analyses and machine learning applications.

Given these challenges and the critical information contained within fluorescent images, there is a compelling need to augment scarce datasets with high-fidelity synthetic images. Generative models, particularly those that are diffusion-based, offer a promising avenue for synthesizing realistic fluorescent cell images that preserve essential anatomical and functional details while mimicking the characteristics found in real data. Such augmentation not only helps overcome data scarcity but also enhances the training of AI models deployed in segmentation, classification, and anomaly detection tasks. The potential to reliably generate synthetic images that replicate the nuanced features of real fluorescent cell data stands to significantly improve both experimental reproducibility and diagnostic accuracy.

3.5 Applications of my Research in Neuroscience

The evolution of neuroscience imaging has played a pivotal role in advancing our understanding of the brain, from the early histological techniques developed by Golgi and Cajal to the sophisticated fluorescent and electron microscopy methods used today. Early breakthroughs in neuroanatomy relied on staining techniques to reveal neuronal structure, laying the groundwork for modern imaging approaches. The introduction of electron microscopy in the mid-20th century enabled visualization of finer neuronal structures, though its reliance on fixed tissue limited its ability to capture dynamic processes. Later, the emergence of fluorescence microscopy revolutionized the field by allowing researchers to simultaneously label specific proteins or organelles, thereby revealing details of synaptic connectivity and cellular dynamics in live tissue. Techniques such as confocal and two-photon microscopy have subsequently enhanced spatial resolution and penetration depth while facilitating in vivo studies. However, despite these advances, fluorescence imaging remains constrained by challenges such as photobleaching, phototoxicity, and autofluorescence, all of which contribute to data variability and the limited availability of high-quality datasets [20].

The book “Portraits of the Mind: Visualizing the Brain from Antiquity to the 21st Century” by Carl Schoonover is the inspiration behind my selection of this topic, with my hopes being to lay groundwork for future research in this domain. As such, I recommend any interested reader to read the book.

It is, however, important to note that I did not use images of neurons in the training of my model, as these images typically contain a higher number of channels. The multi-channel nature of neuronal images adds a layer of complexity to both the data preprocessing and model training stages, which could not only hinder the development and obscure the performance improvements in my model but were also beyond the scope of this project. Instead, by focusing on fluorescent cell images with three channels, I aimed to simplify the analysis while still capturing the essential subcellular details necessary for understanding cellular processes, whilst laying the foundation for future work in this field.

3.6 Issues of Small Datasets

Another persistent challenge in biomedical imaging, and particularly in fluorescent cell imaging that I aimed to solve, is the limited availability of fluorescent cell image datasets. Unlike natural image datasets such as ImageNet, which contain millions of labelled images [21], datasets in biomedical domains are often orders of magnitude smaller. For example, publicly available datasets such as the Fluorescent Neuronal Cells dataset on Kaggle comprise only a few hundred images [22]. In some cases, even when larger datasets are available, they may lack comprehensive annotations or suffer from variability in imaging protocols.

The limited dataset size poses several challenges for training deep learning models. Deep neural networks, by their very nature, require large amounts of data to learn robust features and avoid overfitting. When datasets are small, models are prone to memorizing training examples rather than learning generalizable representations. This is exacerbated in cases where the imaging data is noisy or where the variability in cell morphology is high.

To mitigate these issues, researchers have employed several strategies. Data augmentation techniques, such as rotations, flips, and intensity scaling, are commonly used to artificially increase the size of the training dataset [23]. While these methods can help, they may not fully capture the variability inherent in biological data.

3.7 Explainability in Medical AI

In recent years, there has been a growing recognition of the need for explainability in AI models, especially in high-stakes fields such as medical imaging. Explainable AI (XAI) refers to techniques and methods that make the decision-making process of AI models transparent and understandable to humans. In the context of diffusion models, explainability is particularly challenging due to the complexity of the underlying stochastic processes and the iterative nature of the reverse diffusion mechanism.

Several recent studies have attempted to provide insights into the internal workings of diffusion models. For instance, research has focused on visualizing the noise removal process at various timesteps to illustrate how the model gradually reconstructs the image [24]. Such visual explanations can help build trust in the model by demonstrating that the synthetic images retain biologically meaningful structures.

In this principle, I aimed to develop my model by closely following the theory taught, breaking it down such that each concept is related to a specific section in my code.

Explainability is not only a technical requirement but also an ethical imperative in biomedical applications. Clinicians and researchers must be able to understand and validate the outputs of AI systems before integrating them into decision-making processes. In cases where AI models are used to generate synthetic data for training diagnostic tools, a lack of transparency can lead to scepticism and hinder clinical adoption. Therefore, the development of diffusion models for fluorescent cell imaging and robust explainability frameworks must occur in parallel such that they

elucidate the reasoning behind generated images and highlight potential limitations or biases in the model.

4. Legal, Social, Ethical and Professional Issues

The debate over AI and its potential impact on humanity's future is extensive and often well-founded. Despite their promising utility, such models carry substantial legal, social, ethical, and professional concerns. In this section, I discuss these concerns, highlighting my approaches and implemented safeguards, as well as important future considerations, to ensure the safe development of diffusion-generated cell imagery for biomedical research practice.

4.1 Legal Considerations

AI regulations vary between jurisdictions. In the UK, “there is currently no legislation in place specifically governing the use of AI in the UK. Instead, AI's use is regulated by a patchwork of more general legislation, such as the UK Medical Device Regulations 2002 or the Data Protection Act 2018, covering certain uses of AI.” [25]. The UK government also declined to sign an international declaration that aims to promote “AI accessibility, and ensuring the tech's development is "transparent", "safe" as well as "secure and trustworthy".” A spokesperson for the government stated this decision was made due to a lack of clarity on global governance and concerns around national security [26].

In contrast, the EU AI Act sets out to specifically categorise AI used in healthcare settings as a high risk system, requiring providers to comply with several procedures, including risk management system, data governance, record-keeping and instructions for use [27]. The act does recognise the importance of innovation, freedom of science and ensuring that advancements in this area continue unimpeded. Therefore, it excludes “AI systems and models specifically developed and put into service for the sole purpose of scientific research and development.” Regardless, the act does require any such activity to be “carried out in accordance with recognised ethical and professional standards for scientific research and should be conducted in accordance with applicable Union law.” [28].

Due to the lack of comprehensive AI-specific legislation in the UK, it is difficult to pinpoint where on the legal spectrum my project lies. Nevertheless, it can be reasonably assumed that, providing strict adherence to ethical guidelines, compliance with copyright regulations in the use of any data, and a full disclosure of any and all potential liabilities and risks to persons should mitigate legal concerns regarding the deployment of this model. My work has been undertaken with a vigilant adherence to all relevant legal and ethical standards. It is also important to note that, at its current state, the model remains a prototype, and as such is not ready for deployment in active research settings.

4.2 Social Considerations

Whilst in its current form the model is not yet viable for research applications and thus poses a limited cause for societal concern, its future social implications warrant careful and thorough consideration.

For researchers, it is crucial that the model not only generates images that are biologically accurate and indistinguishable from the authentic data on which it was trained, but also maintains a high degree of reproducibility, consistency, and robustness. Any deviation from these standards risks introducing subtle errors that could compromise experimental results, misinform subsequent data analysis, or ultimately undermine the credibility of scientific findings. Furthermore, the adoption of such models in research settings requires rigorous validation against established benchmarks to ensure that the synthesized images truly reflect underlying biological structures and processes. Therefore, the development of clear performance metrics and reproducibility tests is essential.

For the broader public, the acceptance of AI-generated biomedical imagery fundamentally depends on the principles of transparency, explainability, and robust assurances of privacy and anonymity [29]. Without clear and explicit communication regarding the synthetic nature of these images, there is a risk that public trust in scientific outputs may erode, particularly if these images are perceived as deceptive or unsafe. Therefore, while it is imperative to meet stringent scientific standards within the research community, it is equally important to ensure that the public remains well informed and assured that ethical guidelines and technical safeguards are rigorously implemented during the evolution of these models.

4.3 Ethical Considerations

Ethics were a key consideration during the development of the model. Great care was taken to ensure that the choice of dataset ensured patient confidentiality and that I followed responsible use guidelines. Additionally, the risk of introducing biases during training was mitigated by ensuring the dataset itself was free of bias, and random transforms were applied to the images to ensure the model learnt in a probabilistic fashion. Whilst this constitutes future work, I also have an ethical responsibility to ensure that the model is not considered fit for use unless objective measures are developed that guarantee the biological accuracy of generated images, as any inaccuracies could lead to potentially harmful outcomes.

4.4 Professional Considerations

Professional considerations during the project adhered strictly to UK standards, including guidelines from the British Computer Society (BCS) Code of Conduct [30] and UK Research and Innovation (UKRI) [31]. Emphasis was placed on scientific rigor, transparency and reproducibility, aligning closely with best practices outlined by the Alan Turing Institute on ethical AI research dissemination [32].

5. System Requirements

In this section I describe the hardware requirements for the project and then discuss both the functional and non-functional requirements.

5.1 Hardware Requirements

Although the PyTorch website does not specify an exact list of minimum hardware requirements, it is reasonable to assume that any system capable of executing Python and equipped with a 64-bit CPU will suffice [33].

Building and training a diffusion model, however, requires significantly powerful hardware. For example, my personal computer is outfitted with an Intel Core i7 9th Gen and a Nvidia GTX 1660Ti GPU. These specifications, while adequate for prototyping, pale in comparison to the free version of Google Colab, which provides access to a Tesla T4 GPU. It is important to note that the use of Google Colab is subject to a variable time limit on usage, with an average of about three hours per day which introduced project scheduling constraints.

5.2 Functional Requirements

The primary function of the model is to generate biologically accurate images of fluorescently dyed cells. As the implementation of an objective quality checker was outside the scope of this project, I relied on human evaluation as a suitable alternative. Ultimately, this functional requirement forms the core objective of the project, driving the selection of data preprocessing techniques, network architecture, and the diffusion framework itself.

5.3 Non-Functional Requirements

Beyond the central functional requirement, the model must also achieve certain non-functional requirements which are:

1. Performance - The model must train and generate images within acceptable time limits despite variable factors such as diffusion steps, epochs, and dataset size.
2. Consistency - The training process should generate reliable and reproducible outputs across different runs, ensuring stability even with the variability imposed by Google Colab.
3. Usability - The model should be straightforward to deploy and adjust, enabling easy integration of new data or modifications to model parameters.
4. Reliability - Measures such as gradient clipping and regular validation checks should ensure that the model behaves predictably over extended training periods.
5. Reproducibility - It is essential that the model's performance can be consistently replicated under the same conditions to support rigorous scientific research.

This comprehensive evaluation of both hardware capabilities and system requirements is essential for understanding the current limitations of the model and for guiding future improvements in the synthesis of biologically accurate fluorescent cell images

6. Design and Implementation

To develop my model, I opted for PyTorch over other frameworks because it provided a straightforward, Python coding environment and an extensive, mature ecosystem that greatly facilitated the development of my diffusion model. PyTorch's intuitive syntax and well-designed data handling libraries such as `torchvision` and `torch.utils.data` made it easier to implement and iterate on my model.

In comparison, TensorFlow often requires more involved configuration and the management of static graphs, which can complicate the development process, particularly when rapid prototyping is desired. Meanwhile, JAX offers impressive performance benefits and advanced automatic differentiation; however, its functional programming paradigm and currently less mature ecosystem would have presented me with a steeper learning curve. The extensive community support available for PyTorch and the abundance of practical examples and documentation further reinforced its suitability for my needs.

Throughout the course of the development, I referred to a variety of online sources including GitHub repositories such as [34], [35], [36], [37], [38], Hugging face tutorial [39], LearnOpenCV [40] and visual explanations on YouTube like [41] and [42].

To describe the design and implementation of my model, I have divided the process into six distinct sections, as outlined below.

6.1 Dataset and Transforms

The dataset comprised of fluorescent microscopy images of stained cancer cells, which were obtained from the EBI BioStudies database under S-BIAD847 which comprises of high-resolution fluorescent microscopy images of stained cancer cells [43]. These images were obtained using advanced multichannel fluorescence techniques designed to capture critical cellular components while minimising common artefacts such as photobleaching and autofluorescence. The imaging protocols for this dataset ensure consistent staining intensity and contrast, enabling the clear visualisation of subcellular structures which are crucial for oncological analysis, such as nuclear morphology and cytoplasmic distributions. In addition, the dataset's quality control measures guarantee that the images contain a high signal-to-noise ratio, thereby preserving the fine details necessary for the accurate interpretation of complex cell characteristics [44].

Initially, the downloaded files were provided as separate channels in greyscale. To prepare these images for model training, I wrote a custom Python script to colourise and merge the channel-separated images into composite colour images that accurately represented the original biological signals.

Once the images had been combined, I applied the `RandomCrop(256)` transform from the `torchvision.transforms` library. This transformation was selected to introduce variability within the latent space while preserving the essential cellular morphology. By randomly cropping sections of each image, the transform not only introduced spatial variability but also effectively increased the size of the training dataset, thereby helping to minimise overfitting.

After performing the random crop, all images were converted into tensors using the `ToTensor()` class from the same library, ensuring that the data was in the appropriate format for processing by PyTorch. Finally, the prepared images were preloaded using the `DataLoader` class from the `torch.utils.data` module, establishing an efficient data pipeline for the training process.

This preprocessing pipeline was crucial for standardising the images and augmenting the dataset, which facilitated the effective training of the diffusion model.

6.2 UNet Architecture

The denoising backbone was implemented using the `UNet2DModel` class from Hugging Face's `diffusers` library. The model operates on RGB input and output `(in_channels=out_channels=3)` with a configurable spatial resolution `(sample_size)`. The network was designed with 6 downsampling and 6 upsampling stages, and 2 residual layers per block.

The `block_out_channels` were set as (64,64,128,128,256,256). This defines the number of output channels at each depth level of the U-Net, progressively increasing the feature dimensionality as the spatial resolution is reduced via convolution. The `down_block_types` and `up_block_types` specify the structure at each level. In particular:

- Downsampling path:
    - Uses a combination of `DownBlock2D` and `AttnDownBlock2D`, introducing spatial attention in the deeper layers to better capture long-range dependencies in cell morphology.
- Upsampling path:
    - Mirrors the downsampling path, with corresponding `UpBlock2D` and `AttnUpBlock2D` modules, including skip connections that preserve fine-grained image features from earlier layers, a trait inherent to UNet models.

Gradient checkpointing was enabled via `enable_gradient_checkpointing()`, allowing intermediate activations to be recomputed during the backward pass to reduce memory usage. This was essential for training stability and feasibility when using limited GPU resources.

The model was moved to the GPU using `.to(device)`

6.3 Cosine-Beta Schedule

Here, β refers to the amount of noise added to an image and α refers to the amount of information remaining in the image after a certain amount of noise has been added.

The noise schedule β_t was implemented using a cosine-based method [3] designed to smoothly vary the noise level across diffusion timesteps. This approach improves training stability and output image quality compared to linear or quadratic schedules.

A 1D tensor of $T + 1$ evenly spaced values from $0\ to\ T$ was generated using `torch.linspace`. The cumulative product of alpha values, $\overline{\alpha}_t$, was calculated using the equation:

$$\overline{\alpha}_t = \left[\cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)\right]^2$$

where $s = 0.0008$ is a small offset added to prevent the function from starting at zero, causing early timesteps to be overly noisy. The $\overline{\alpha}_t$ values were then normalised by dividing by $\overline{\alpha}_0$.

I then calculated the beta values using the formula:

$$\beta_t = 1 - \frac{\overline{\alpha}_t}{\overline{\alpha}_t - 1}$$

These values were then clamped between 0 and 0.999 to prevent issues during training.

From these betas, $\alpha_t = 1 - \beta_t$ was computed, and finally the posterior variance was computed as:

$$\text{posterior_variance}_t = \beta_t \cdot \frac{1 - \overline{\alpha}_{t-1}}{1 - \overline{\alpha}_t}$$

All these tensors - betas, alphas, cumulative products, and posterior variances - were moved to the GPU at the start of training to avoid redundant computations and improve runtime efficiency.

6.4 Forward Process

The forward diffusion process gradually adds Gaussian noise to a clean image x_0 across a fixed number of timesteps $t \in [1, T]$, producing a noisy image x_t. This is implemented using the closed-form solution of the forward process formula defined in [2]. This is obtained as $\alpha = 1 - \beta$ and this variation allows me to directly obtain x_t for any timestep t:

$$x_t = \sqrt{\overline{\alpha}_t} \cdot x_0 + \sqrt{1 - \overline{\alpha}_t} \cdot \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, I)$$

Here:

- $\overline{\alpha}_t$ is the cumulative product of alphas up to timestep t.
- ϵ is standard Gaussian noise.
- x_t is the resulting noisy image.

In my implementation, for each sample in a training batch the following occurs:

- The timestep t is randomly chosen and cast to a `LongTensor`.
- $\overline{\alpha}_t$ is extracted using `alphas_cumprod[t]` and reshaped to match the batch dimensions.
- Its square root and the square root of its inverse are computed.
- A noise tensor ϵ of the same shape as x_0 is generated using `torch.randn_like`.
- The noisy image x_t is produced by scaling and combining x_0 and ϵ.

This forward process is used during training to generate noisy inputs, that the model then uses along with ϵ to learn to predict the amount of noise added to x_t.

6.5 Reverse Process

The reverse process reconstructs a clean image $\hat{x}_0$ from Gaussian noise $x_T \sim \mathcal{N}(0, I)$ by iteratively denoising across T timesteps using the learned model. The sampling is performed in reverse order, from $t = T$ down to 1, with each step generating x_{t-1} from x_t.

At each timestep t, the model predicts the added noise $\epsilon_\theta(x_t, t)$. The predicted clean image $\hat{x}_0$ is then computed as:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}} \cdot \epsilon_\theta(x_t, t)\right) + \sigma_t z$$

If $t > 0$, a small amount of noise z, scaled by σ_t is added to the image. This is because, intuitively, diffusion models are deterministic. To avoid predictable outputs, a small amount of Gaussian noise is reintroduced to the image, forcing the model to sample from the wider latent space, thereby forcing stochasticity. For $t = 0$, the sampling stops, and the image is returned. To ensure numerical stability, all scalar coefficients are precomputed and reshaped to match the dimensions of the image tensors. Sampling is performed under inference conditions, with gradients disabled and the model in evaluation mode.

This iterative denoising process enables the transformation of a purely stochastic input into a possibly coherent, biologically plausible fluorescent cell image, highlighting the model's ability to capture and reconstruct spatial patterns learned from the training distribution.

6.6 Training Loop

The model was trained to minimise the standard denoising diffusion loss, where the objective is to predict the noise ϵ added to a clean image x_0 at a randomly sampled timestep t. At each iteration, a batch of images was drawn from the training dataset, and the corresponding noisy samples x_t were generated using the forward diffusion process. The model was then optimised to minimise the mean squared error between the predicted and true noise.

The training loop was implemented in PyTorch using mixed precision with `torch.cuda.amp` to accelerate training and reduce GPU memory consumption. Gradients were scaled and backpropagated through the network using `GradScaler`, and gradient clipping with a maximum norm of 1.0 was applied to improve numerical stability.

The model was trained over a fixed number of epochs using the Adam optimiser. During training, losses were logged periodically, and the model was evaluated qualitatively by sampling images from noise every 20 steps. These samples were visualised using Matplotlib and saved to disk for monitoring the progression of generation quality.

Model checkpoints were saved at the end of each epoch, ensuring recoverability and enabling retrospective analysis of model performance at various stages of training. The training procedure was carried out with the model in `.train()` mode, and sampling was performed with gradients disabled and the model set to `.eval()`.

7. Testing and Success Measurement

Qualitative evaluation played a central role in assessing model performance. Generated images were visually inspected throughout training to observe and track the learning progress of the model. While inherently subjective, visual inspection remains a practical and widely used method in generative modelling, particularly for biomedical imagery where structural integrity and realism are critical.

To support this subjective analysis, quantitative testing was conducted using the Mean Squared Error loss function between the predicted noise and the true noise added as an indicator of model performance. A consistent decline in this value across epochs indicated that the model was successfully learning the denoising objective. Due to time, scope and hardware constraints, more advanced metrics such as Fréchet Inception Distance (FID) or Structural Similarity Index (SSIM) were not implemented, but their use would be the logical next step in evaluating output quality.

Practical limitations were also considered in the evaluation of success. Due to resource constraints associated with running the model on Google Colab, issues such as session interruptions and memory crashes served as indirect indicators of model complexity and training demands. This required me to explore unique ways of optimising the model, something that inadvertently brought down training times, although marginally.

8. Project Management

The project loosely followed a milestone-based development approach, structured around the following four key stages:

1. Model Development - Designing and implementing a working diffusion model capable of generating fluorescent cancer cell images.
2. Initial Training and Validation - Training the model to a baseline level of performance, ensuring it could produce coherent outputs and converge under standard conditions.
3. Optimisation and Demonstration - Improving model output quality through hyperparameter tuning and architectural adjustments, followed by demonstrating its capabilities to my inspector.
4. Final Write-Up - Documenting the methodology, results, and reflections in the final report.

Due to the iterative and experimental nature of machine learning workflows, an agile and flexible development style was adopted. Regular qualitative assessments and discussions with my supervisor informed decisions on when to shift focus between training, evaluation, and refinement.

While formal software engineering methodologies were not strictly applied, time and resource constraints were accounted for by regularly reviewing progress and adapting priorities, particularly when working within the limits of Google Colab.

9. Evaluation

This section evaluates how well my model meets the previously stated functional and non-functional requirements by examining both the training data and the images generated by the model. Functionally, the primary goal was to produce biologically relevant fluorescent cell images that reflect the patterns and structures observed in the real dataset. From a non-functional point-of-view, the focus was on the model's performance, consistency, usability, reliability and reproducibility.

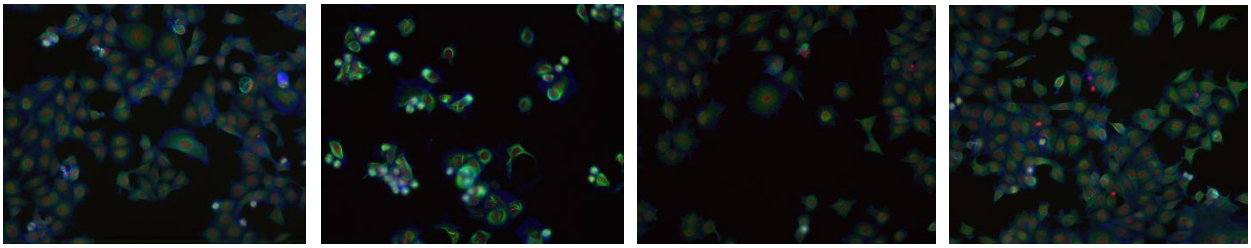

Fig. 2. Samples from the real training data

To illustrate the model's progress, Figure 2 shows a set of 4 images chosen randomly from the set of real training images, whereas Figure 3 shows a set of 9 images selected to display the evolution of the model's learning progress.

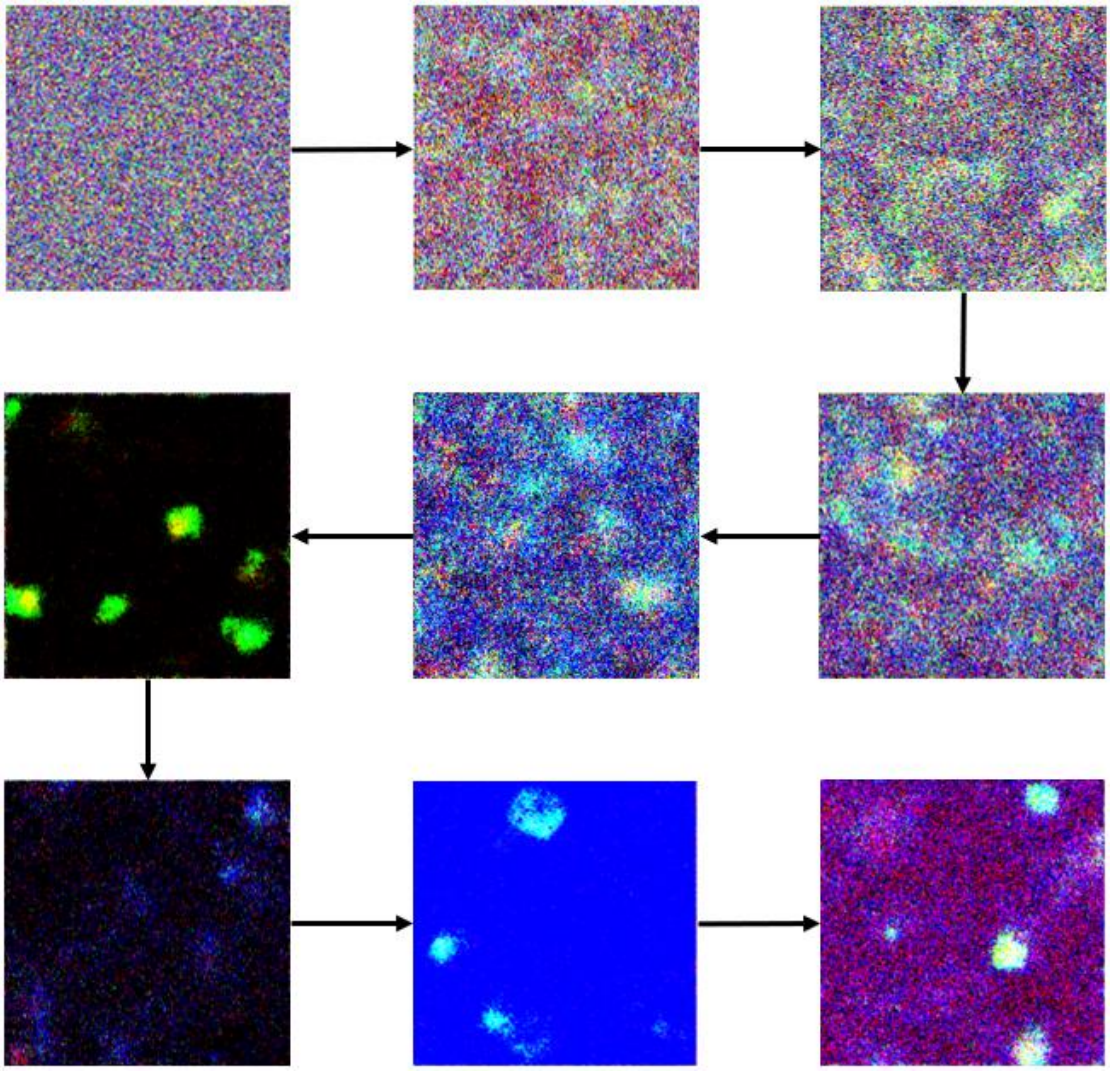

Fig. 3. Images generated during the training of the model, showing the evolution of learning.

9.1 Achievements

Despite the limitations explained in the section below, the model demonstrated key achievements in the design and implementation of a DDPM.

Firstly, a DDPM was purpose built and trained from scratch rather than making use of pre-developed options. This ensured that the model was tailored to the functional and non-functional requirements highlighted previously. It also laid a flexible foundation for future work.

Secondly, the model was trained to the point where it began to generate images reminiscent of the training data. While imperfect, these images displayed the learning of underlying cell morphology and highlight the model's potential to generate images indifferentiable from the training data.

Finally, the project gave due consideration to ethical, legal, social and professional concerns, highlighting their importance and comparing the model's development to these principles. Additionally, the importance of learning materials accessible to beginners has been emphasised, calling for simplified explanations that are approached from a different perspective.

9.2 Limitations

Noting the achievements, there remain many limitations in my model with regards to the functional and non-functional requirements.

Firstly, the core functional requirement for the model to generate biologically accurate images of fluorescent cells was not achieved completely. Whilst preliminary results demonstrate the model's learning of cellular morphology, closer inspection reveals various artefacts and inconsistencies. Some regions appear overly saturated or pixelated, and the boundaries between cells are often ill-defined, indicating that the model has not yet fully captured the complexities of real fluorescent microscopy data. Additionally, the colour channels differ in ways that do not accurately reflect the biological signals seen in the training images. These issues point to the need for further refinements, a few of which are calibrating the noise schedule, tuning hyperparameters, or introducing more robust regularisation strategies.

Secondly, the reliance on human evaluation to assess image quality represents a significant limitation in the validation process. An automated and objective measure of biological accuracy was beyond the project's scope, but its absence means the functional requirement of reliable accuracy assessment remains unmet. Without such a metric, it is difficult to determine quantitatively how closely the synthetic images match the real data distribution.

From the perspective of non-functional requirements, several issues also emerged:

- The performance, reliability and consistency of my model were affected by the limitations of Colab's variable GPU access time restrictions, which prevented me from running extended training sessions and image generation tests. These would have allowed me to further improve and optimise the model.
- In its development-focused state, the model is not accessible to beginners, as it lacks a user-friendly interface, detailed documentation and improved modularity that would enable researchers to swap and implement features with ease.
- Since the model is yet to learn the finer details from the training data and generate accurate images, synthesised images display stochasticity, suggesting there exist gaps in the latent space.

Lastly, the exclusion of multi-channel data due to computational and design constraints limited the model's applicability to real-world biomedical imaging tasks, which often rely on richer, multi-

dimensional datasets. This also affects the non-functional requirement of extensibility, as adapting the model to more complex biomedical inputs would require substantial additional work.

Overall, despite the seemingly limited progress, the model's progression thus far suggests that it holds promise for generating realistic fluorescent cell images in the future. Enhanced training methods, larger datasets, or domain-specific objective metrics could help refine the synthetic outputs and bring them closer to biological fidelity. Ultimately, while the model meets some of the initial functional requirements by learning salient features of the data, improving image quality and ensuring consistency over multiple training runs remain key steps before it can be suitable for more advanced research applications.

10. Future Work

While this project has demonstrated the potential for DDPMs to generate synthetic fluorescent cell images with high fidelity, several areas remain for further research and development. One promising direction involves refining the noise schedule and optimisation strategies to enhance model stability and training efficiency. For instance, investigating adaptive noise scheduling mechanisms that would dynamically adjust noise levels based on intermediate reconstruction quality could further improve the trade-off between detail preservation and computational cost.

Another area for future exploration is the development of objective evaluation metrics tailored to biomedical imaging. Currently, human evaluation is the only method of validating the quality of images generated by my model. However, an automated, quantitative measure of biological accuracy is essential for broader adoption in clinical research. This would involve integrating domain-specific metrics such as quantitative comparisons of morphological features or biological accuracy and utilising these metrics to fine-tune the diffusion process.

Expanding the scope of the model to incorporate multi-channel data and multimodal imaging presents an additional avenue for research with the greatest potential for positive impact. While this project focused on fluorescent cell images with three channels to simplify the model, many applications in neuroscience and medicine require processing complex multi-channel images to capture a wider array of biological signals. Future works could extend the framework to accommodate the higher-dimensional data inherent in whole neuron imaging or even multimodal datasets that combine fluorescence microscopy with other imaging modalities, such as magnetic resonance imaging.

Moreover, the integration of explainability techniques within the diffusion framework could help demystify the generative process. By developing methods that trace which features in the model and training data contribute most significantly to the final output, researchers can provide clinicians and scientists with insights into the model's decision-making, thereby enhancing trust and facilitating model adoption in clinical workflows.

Finally, future work in fine-tuning the diffusion model with task-specific objectives such as generating images with known pathologies or cell types, could open possibilities for more targeted applications in diagnostics or personalised medicine.

11. Conclusion

This project has demonstrated the potential of using diffusion models for the generation of synthetic fluorescent cell images and offers promising avenues for future research in biomedical imaging and neuroscience. My work has shown that, despite the imperfections currently observable in the generated images, the model is beginning to learn and replicate key morphological patterns inherent to the training data.

Throughout this project, emphasis was placed on ethical considerations, rigorous data preprocessing, and comprehensive research and development. The training dataset, comprising high-resolution images from the BioImage Archive, was meticulously processed to preserve the biological integrity of the data while enhancing variability to reduce overfitting. The choice to focus on fluorescent cell images rather than whole neuron images with multiple channels was made to simplify the computational pipeline and to focus on subcellular details that are critical for understanding disease pathology.

While the current model is not yet viable for deployment in research due to the presence of artefacts and a lack of fully objective validation metrics, it nonetheless is a solid foundation for further optimisation. Future work should aim to refine the noise schedule and hyperparameter tuning, develop automated measures for assessing biological fidelity, and possibly explore hybrid models that combine the stability of diffusion processes with the rapid inference capabilities of other generative frameworks. Additionally, advancing explainability remains important, as integrating techniques to elucidate the mechanisms by which the model reconstructs and synthesises complex image features could enhance trust in the technology and provide deeper insights into the generative process. Furthermore, there is a clear need for improved, accessible educational resources on diffusion models to foster broader understanding and innovation in this rapidly evolving field.

In summary, this project highlights both the promise and the challenges inherent in applying diffusion models to biomedical imaging. By addressing these challenges in future work, I believe that such models can eventually provide a robust tool for augmenting biomedical datasets, advancing both diagnostic applications and fundamental research.

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13. Appendices

All the code for the model and the dataset augmentation script can be found at https://git.cs.bham.ac.uk/axa1859/final-year-project-code